Observed Dyness Powerbrick Pro 14.3 kWh BMS and Charge-Control Behaviour

Experimental characterization of passive balancing, indirect channel-current estimation, full-charge detection, CCL control and charge-MOSFET protection

Battery software version 2.50-71.10.11

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Scope. This report documents experimentally observed BMS behaviour. It is not an official Dyness specification and does not replace manufacturer limits or warranty conditions.

Abstract

A Dyness Powerbrick Pro 14.3 kWh battery with a 16-cell BMS running software version 2.50-71.10.11 was experimentally investigated using continuous telemetry of individual cell voltages, battery current, state of charge (SOC), charge-current limit (CCL), charge-permission status, raw status bytes and BMS temperature. Cells 1-15 are transmitted directly; cell 16 is reconstructed in the recorded CSV from pack voltage and the sum of cells 1-15. The measurements reveal three largely independent control layers: passive balancing, communication-level charge control through CCL, and physical cell-overvoltage protection through the charge MOSFET. Passive balancing is globally gated by battery charge current and individual balancing resistors are selected relative to the minimum cell voltage. A separate matched-state telemetry calculation estimates approximately 60 mA per selected cell or active balancing channel between 3.5 V and 3.6 V, corresponding to approximately 58.7-60.7 ohm effective bleed resistance, 0.20-0.22 W per active channel and a conservative 50-70 mA interpretation range. This is an indirect engineering estimate, not a direct resistor-current measurement or manufacturer specification. The BMS declares the battery full when the highest cell reaches 3.5 V, whereupon SOC becomes 100%, CCL becomes 0 A and Charge Enabled becomes false, while the charge MOSFET remains on. If current nevertheless continues, balancing remains active between 3.5 V and 3.6 V. At 3.6 V the charge MOSFET opens and physically interrupts charging; it closes again when discharge is detected; no independent maximum-cell-voltage recovery threshold was established. CCL state and charge-MOSFET state are therefore separate control mechanisms.

Summary of observed behaviour

Function / parameter	Observed criterion	Observed result
Battery software version	2.50-71.10.11	Version under test
Balancer global enable	Ibat >= 1.5 A for 30 s	Passive balancing enabled; selected channels approx. 60 mA each
Balancer global disable	Ibat <= 1.4 A for 30 s	Passive balancing disabled
Individual resistor ON	Vcell - Vmin >= 30 mV	Resistor for that cell enabled
Full detection	Vmax >= 3.5 V	Full status; SOC = 100%; CCL = 0 A; Charge Enabled = false
Normal charge permission restored	SOC 100% -> 99%	CCL = 56 A; Charge Enabled = true
Charge MOSFET OFF	Vmax >= 3.6 V	Physical charging path interrupted
Charge MOSFET ON during discharge after OVP	Discharge detected after cutoff	Physical charging path restored
Balancing at 100% SOC	Actual Ibat >= 1.5 A and cell delta criterion met	Balancing continues despite CCL = 0 A; approx. 60 mA per selected cell
Explicit balancing flag	None identified	No change in decoded status bytes during ON/OFF transitions
Full/high-cell flag	status3: 0x80 -> 0x88	Bit 0x08 set

Key thresholds: 1.5 A / 1.4 A global current gate, approximately 60 mA per selected balancing cell or active channel, 30 mV individual-cell delta, 3.5 V full-detection threshold, and 3.6 V hardware cutoff threshold.

1. Disclaimer and interpretation conventions

The values and algorithms described in this report were derived from experimental observation during controlled charge and discharge tests. They are not taken from Dyness firmware documentation or an official manufacturer specification.

Thresholds were determined from recorded telemetry, individual cell voltages, BMS temperature behaviour, battery current, CCL, status bytes and physical charge-MOSFET behaviour. Where an observed transition occurred close to an obvious nominal threshold, the nominal value is used throughout this report. Thus, transitions measured around 28-30 mV are reported as 30 mV; the full-charge threshold is reported as 3.5 V; and the hardware cutoff is reported as 3.6 V.

The values therefore describe the behaviour of the tested BMS and software version 2.50-71.10.11. Other hardware or software revisions may implement different thresholds, timing, hysteresis or protection behaviour. Positive battery current denotes charging current.

Engineering hypothesis: balancer time qualification

The balancing-current experiments show a delayed and approximately symmetrical response when battery current crosses the balancing enable and disable thresholds. This delay is interpreted as an algorithmic qualification time rather than primarily as thermal sensor lag. The working hypothesis used for further testing is:

```
if Ibat >= 1.5 A continuously for 30 s:
    enable balancing

if Ibat <= 1.4 A continuously for 30 s:
    disable balancing
```

The 30-second timing criterion is an engineering hypothesis inferred from the observations and has not been verified from firmware documentation. It is therefore covered by this disclaimer.

2. Measurement basis and detection method

The analysis used time-series telemetry with approximately 6-second sampling. The CSV represents all 16 physical cells. Cells 1-15 are transmitted directly by the BMS. Cell 16 is reconstructed by the logger as V16 = Vpack - sum(V1...V15) and is stored in the CSV as battery_02_cell_16_v. Other relevant signals include battery current, SOC, CCL, Charge Enabled / Discharge Enabled states, five raw status bytes, ordinary battery temperature sensors and the BMS temperature sensor.

Passive-balancer operation is not explicitly exposed in the decoded status data. Its activity was therefore inferred from changes in internal BMS heat generation under otherwise nearly constant conditions. Stepwise increases in BMS temperature slope were correlated with individual cells crossing candidate voltage-difference thresholds. A separate current sweep around 1.4-1.5 A was used to identify the global balancing-current gate; the accompanying matched-state calculation estimates the resulting selected-channel current at approximately 60 mA per active cell.

Cell 16 is a valid reconstructed representation of the sixteenth physical cell and is included throughout the recorded dataset. Because it is obtained by subtraction from the pack voltage, its numerical trace inherits pack-voltage quantization and the accumulated quantization of cells 1-15. For identifying millivolt-scale resistor switching thresholds, the directly transmitted cells 1-15 are therefore used as the primary high-resolution reference. The inferred BMS algorithm itself applies to all 16 physical cells.

3. Functional architecture

The observed BMS behaviour is best understood as five distinct functions that share measurements but are not identical control paths:

- Global passive-balancer enable based on actual battery charging current.
- Individual balancing-resistor selection based on each cell relative to the minimum cell.
- Full-charge detection at the highest-cell threshold.
- Communication-level external charge control through CCL and Charge Enabled.
- Physical cell-overvoltage protection through the internal charge MOSFET.

All cell-based state variables in the inferred control model (Vmin, Vmax and individual resistor selection) refer to the complete 16-cell series string. In the measurement data, cells 1-15 are direct telemetry values and cell 16 is reconstructed from pack voltage.

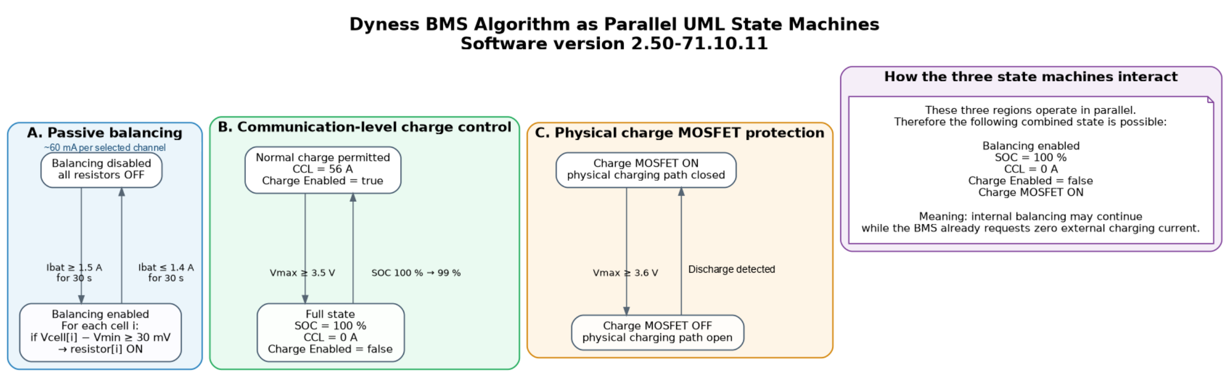

Figure 1. UML state-machine view of the three parallel control layers: passive balancing (approximately 60 mA per selected cell), communication-level charge control and physical charge-MOSFET protection.

```
Battery current -> global balancing gate
Cell delta to Vmin -> individual resistor selection

Vmax = 3.5 V -> Full -> SOC 100% -> CCL 0 A -> Charge Enabled false
                 charge MOSFET remains ON; balancing may continue

Vmax = 3.6 V -> charge MOSFET OFF -> charging physically interrupted
Discharge detected -> charge MOSFET ON -> physical charging path restored
```

4. Global passive-balancer current gate

The passive balancing system is globally enabled only when the battery receives sufficient charging current. Repeated transitions around the threshold produced reproducible changes in internal BMS heat generation. This 1.5 A / 1.4 A gate is the pack-level enable condition; it is distinct from the approximately 60 mA current estimated for each selected balancing channel.

```
Enable threshold:  Ibat >= 1.5 A
Disable threshold: Ibat <= 1.4 A
```

At 1.5 A the BMS temperature rose; at 1.4 A and below it fell; returning to 1.5 A restored heating. The 0.1 A separation constitutes current hysteresis at the telemetry resolution.

4.1 Time-qualified state-machine hypothesis

```
if balancing_enabled == false:
    if Ibat >= 1.5 A continuously for 30 s:
        balancing_enabled = true

if balancing_enabled == true:
    if Ibat <= 1.4 A continuously for 30 s:
        balancing_enabled = false
```

This model explains the nearly symmetrical delay in both directions and avoids balancer chatter when current fluctuates around the threshold.

5. Individual balancing-resistor selection

Once balancing is globally enabled, individual resistors are selected according to each cell voltage relative to the lowest cell in the 16-cell battery. A selected cell is represented by an effective balancing channel drawing approximately 60 mA in the 3.5-3.6 V upper-cell region. Vmin in the inferred algorithm therefore denotes the true minimum of all 16 physical cells. The event-by-event threshold extraction gives greatest weight to cells 1-15 because those channels are transmitted directly, whereas cell 16 is reconstructed from pack voltage.

```
Vmin = minimum(cell voltages)

for each cell i:
    if Vcell[i] - Vmin >= 30 mV:
        balancing resistor[i] = ON
    else:
        balancing resistor[i] = OFF
```

At the detected resistor-activation events, newly selected cells consistently lay about 30 mV above the minimum. Their voltage above the cell average varied substantially, so a fixed threshold relative to average voltage does not reproduce the observed sequence. The minimum cell is the consistent reference.

5.1 Absolute voltage region

During the recorded activation sequence, the relevant cells were around 3.4 V or higher. The data demonstrate balancing under these conditions but do not independently establish a separate absolute balancer-enable voltage below this region. The directly established individual selection criterion is therefore the 30 mV delta relative to Vmin.

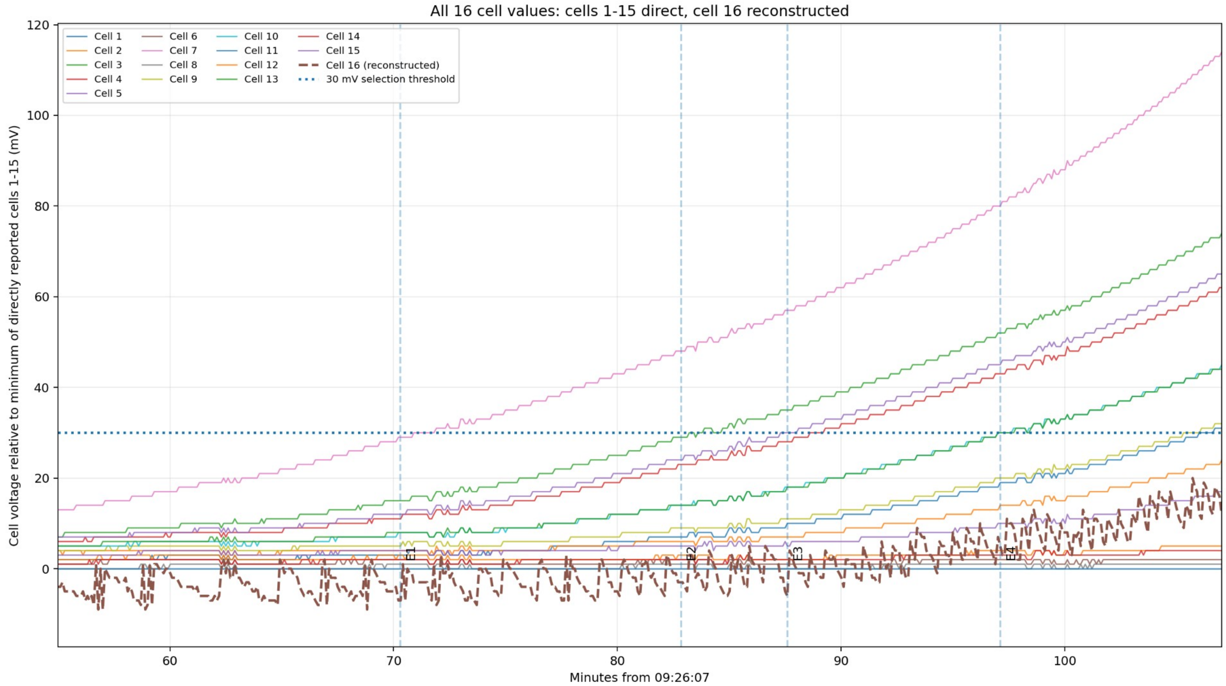

Figure 2. All 16 cell values are represented. Cells 1-15 are direct telemetry; cell 16 is reconstructed from pack voltage and shown separately. The 30 mV selection threshold is evaluated against the high-resolution directly transmitted channels to avoid reconstruction quantization bias.

6. Thermal evidence for passive balancing

No explicit passive-balancing flag was identified in the decoded telemetry. BMS temperature therefore provides the clearest indirect indication of balancing resistor power. Multiple resistor activations appear as stepwise increases in internal heat generation, whereas disabling the global balancing gate produces cooling under otherwise similar conditions. The matched-state calculation estimates approximately 0.20-0.22 W dissipated by each active channel at the upper-cell voltages, consistent with approximately 60 mA per selected cell.

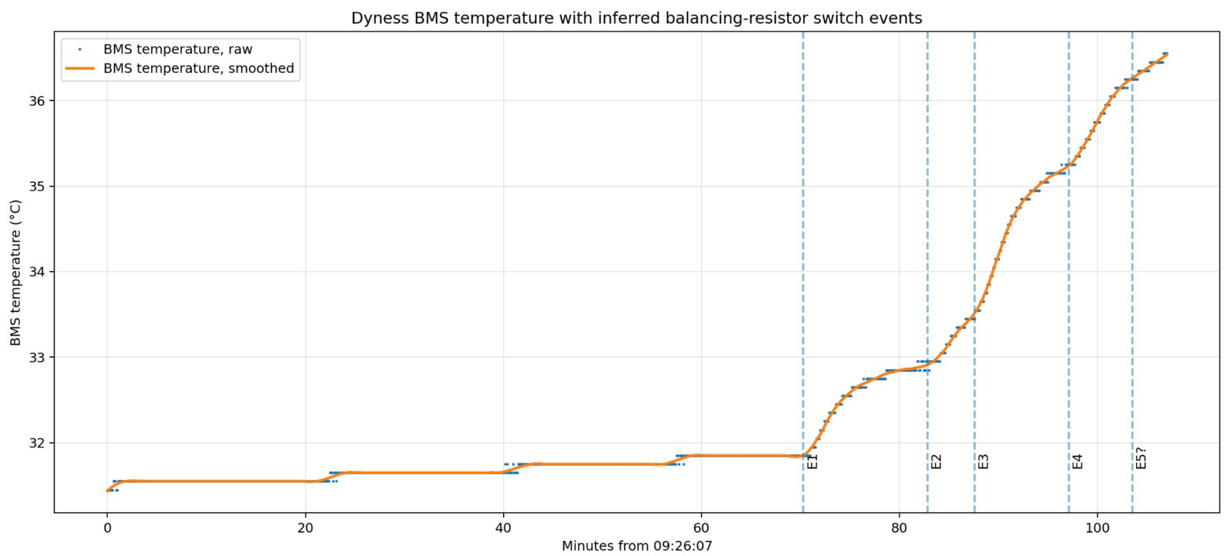

Figure 3. BMS temperature with inferred balancing-resistor switch events during constant low-current charging.

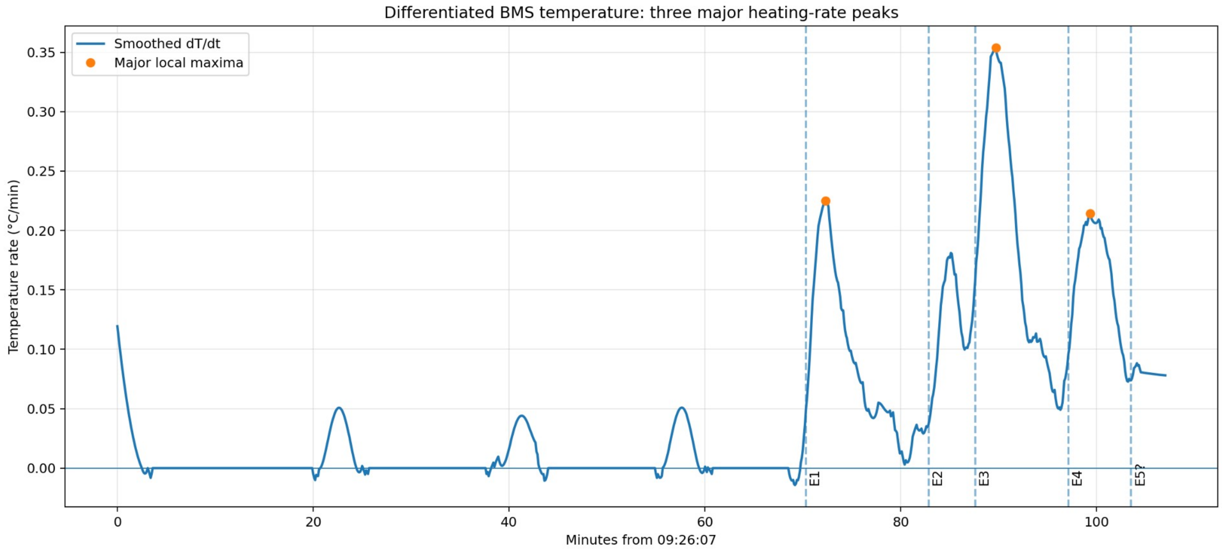

Figure 4. Smoothed derivative of BMS temperature. Local increases in heating rate identify changes in internal dissipation.

The approximately delayed temperature response following the 1.4/1.5 A current transitions is interpreted in this report primarily as the 30-second algorithmic qualification described in Section 1, with any remaining sensor thermal response treated as secondary.

7. Status flags during balancing

The decoded status bytes did not change during current-induced balancing ON/OFF transitions. The observed values remained:

```
status1 = 0x00
status2 = 0x0E
status3 = 0x80
status4 = 0x00
status5 = 0x00
```

SOC, CCL, Charge Enabled and Discharge Enabled also remained unchanged during these balancing transitions. No directly transmitted passive-balancing ON/OFF indicator has therefore been identified in the currently decoded interface.

8. Full-charge detection at 3.5 V

The BMS declares the battery full when the highest individual cell reaches 3.5 V. The transition does not require a balanced pack; a single high cell can force the complete battery to 100% SOC while other cells remain substantially lower.

```
Vmax reaches 3.5 V
    -> Full/high-cell status set
    -> SOC = 100%
    -> CCL = 0 A
    -> Charge Enabled = false
```

In the recorded transition, status3 changed from 0x80 to 0x88. The added bit 0x08 appeared before the subsequent telemetry cycle reported SOC = 100%, CCL = 0 A and Charge Enabled = false. This bit is associated with the observed full/high-cell state and is not a passive-balancing flag.

9. SOC, CCL and communication-level charge permission

The normal observed CCL is 56 A. At 100% SOC the BMS requests zero external charging current. When SOC falls from 100% to 99%, the normal charge request is restored.

```
if SOC == 100%:
    CCL = 0 A
    Charge Enabled = false

if SOC <= 99%:
    CCL = 56 A
    Charge Enabled = true
```

The recovery of communication-level charge permission is therefore controlled by SOC. It is not identical to charge-MOSFET recovery.

10. Charge permission versus physical charge MOSFET

CCL = 0 A and Charge Enabled = false do not open the internal charge MOSFET. The battery remains electrically connected, and current can still flow if the external charger ignores the BMS request. The charge MOSFET is reserved for physical cell-overvoltage protection.

10.1 Charge-MOSFET overvoltage cutoff and discharge recovery

```
if Charge_MOSFET == ON:
    if Vmax >= 3.6 V:
        Charge_MOSFET = OFF
        charging current = physically interrupted

if Charge_MOSFET == OFF:
    if discharge is detected:
        Charge_MOSFET = ON
        physical charging path = restored
```

The physical protection path switches the charge MOSFET off at the observed 3.6 V overvoltage cutoff. After the cutoff, the MOSFET closes again when discharge is detected; the measurements do not establish an independent maximum-cell-voltage recovery threshold.

10.2 Re-enabling the MOSFET does not restore CCL

After an overvoltage cutoff, discharge detection restores the physical charging path, but the BMS can still remain at SOC = 100%, CCL = 0 A and Charge Enabled = false. CCL returns to 56 A only after SOC falls to 99%. This demonstrates that the charge MOSFET and communication-level charge permission are independent mechanisms.

11. Balancing in the 3.5-3.6 V region

Passive balancing continues after full detection. The BMS can therefore simultaneously report SOC = 100%, CCL = 0 A and Charge Enabled = false while the charge MOSFET remains on and balancing resistors remain active at approximately 60 mA per selected cell.

```
3.5 V <= Vmax < 3.6 V

SOC = 100%
CCL = 0 A
Charge Enabled = false
Charge MOSFET = ON
Passive balancing can remain active
```

If actual battery current remains at or above the balancing-current threshold, passive balancing can continue according to the 30 mV cell-selection criterion. Each selected channel is estimated to draw approximately 60 mA in this region. This operating region is technically accessible because the physical charging path is still closed, but it is outside the BMS-authorized external charging condition.

The continued operation of the internal balancer does not constitute permission for an external charger to ignore CCL = 0 A. Deliberate charging in this region can conflict with the BMS command and may have manufacturer warranty implications. The 3.6 V threshold is a hardware protection threshold, not a normal charge target.

12. Observed operating states

State	Entry criterion	Result
Balancer disabled	Ibat <= 1.4 A for 30 s	Balancing OFF
Balancer qualification	Ibat >= 1.5 A for 30 s	Global balancing enabled
Normal charge with balancing	SOC <= 99%; Vmax < 3.5 V; Ibat >= 1.5 A	CCL 56 A; Charge Enabled true; MOSFET ON; ~60 mA/cell

State	Entry criterion	Result
Full state	Vmax reaches 3.5 V	SOC 100%; CCL 0 A; Charge Enabled false; MOSFET ON; balancing can continue at approx. 60 mA per selected cell
Charge-prohibited but physically connected	3.5 V <= Vmax < 3.6 V	CCL 0 A; MOSFET ON; balancing can continue at approx. 60 mA per selected cell; external current can physically flow
Hardware overvoltage cutoff	Vmax >= 3.6 V	Charge MOSFET OFF; charging physically interrupted
Hardware recovery	Discharge detected after cutoff	Charge MOSFET ON; physical path restored; CCL may remain 0 A
Normal charge permission restored	SOC 100% -> 99%	CCL 56 A; Charge Enabled true

13. Consolidated observed control model

```
Vmin = minimum(cell voltages)
Vmax = maximum(cell voltages)

# Global passive-balancer gate
if balancing_enabled == false:
    if Ibat >= 1.5 A continuously for 30 s:
        balancing_enabled = true

if balancing_enabled == true:
    if Ibat <= 1.4 A continuously for 30 s:
        balancing_enabled = false

# Individual balancing-resistor selection
# Effective selected-channel current: approximately 60 mA per cell
if balancing_enabled:
    for each cell:
        if Vcell - Vmin >= 30 mV:
            balancing resistor = ON
            effective balancing current ~= 60 mA
        else:
            balancing resistor = OFF
else:
    all balancing resistors = OFF

# Full detection / external charge command
if Vmax >= 3.5 V:
    Full status = true
    SOC = 100%
    CCL = 0 A
    Charge Enabled = false
    # Charge MOSFET remains ON.
    # Passive balancing may continue.

# Normal communication-level charge permission
if SOC <= 99%:
    CCL = 56 A
    Charge Enabled = true

# Physical cell-overvoltage protection
if Charge_MOSFET == ON and Vmax >= 3.6 V:
    Charge_MOSFET = OFF
    charging current = physically interrupted

# Charge-MOSFET recovery
if Charge_MOSFET == OFF and discharge is detected:
    Charge_MOSFET = ON
    physical charging path = restored

# Important:
# Charge_MOSFET = ON does not imply CCL > 0.
```

14. Engineering interpretation

The measurements support a layered BMS architecture. Passive balancing is an internal energy-dissipation function controlled by actual current and cell imbalance; the associated matched-state calculation estimates approximately 60 mA per active channel, with approximately 0.20-0.22 W dissipation per channel. Normal charging is controlled externally through CCL. Physical disconnection is reserved for a higher cell-overvoltage threshold. The channel-current estimate is indirect and should not be read as a manufacturer specification.

```
Passive balancing:
    Ibat current gate + 30 mV cell-selection rule
    approximately 60 mA per selected cell / active channel

Normal external charge control:
    Vmax = 3.5 V -> SOC 100% -> CCL 0 A

Emergency physical protection:
    Vmax = 3.6 V -> charge MOSFET OFF

Physical recovery:
    Discharge detected -> charge MOSFET ON

Communication-level recovery:
    SOC 100% -> 99% -> CCL 56 A
```

The most important architectural consequence is that CCL and the charge MOSFET must not be treated as the same signal. At 3.5 V the BMS requests that the external charger stop but deliberately leaves the physical path connected. At 3.6 V it physically disconnects charging. When discharge is detected, it restores the physical path even if CCL is still 0 A.

For normal operation, CCL = 0 A must be respected as the BMS charge-stop command. The 3.6 V charge-MOSFET threshold is a protection limit and must not be used as a normal charging target.

15. Limits of the present observations

- The 30-second balancer qualification time is an engineering hypothesis inferred from the observed delay and should be verified with deliberately timed current pulses. The approximately 60 mA per-channel value is an indirect matched-state estimate with a conservative 50-70 mA interpretation range; no direct resistor-current probe or calorimetric calibration was performed.
- The data demonstrate balancer operation around 3.4 V and above but do not independently establish a lower absolute cell-voltage gate for balancing.
- No explicit passive-balancing flag has been identified in the decoded status bytes; thermal behaviour remains an indirect indicator.
- The reported thresholds characterize the tested battery and software version 2.50-71.10.11 and should not be generalized to other revisions without verification.
- The CSV contains all 16 cell values. Cells 1-15 are directly transmitted; cell 16 is reconstructed as Vpack - sum(V1...V15). Reconstruction introduces additional quantization, so cells 1-15 are preferred for millivolt-scale transition fitting while cell 16 remains part of the complete 16-cell pack model. The calculation source for the current estimate is dyness-balancer-2026-08-09T11-28-32(3).csv, containing 3,207 rows and 62 columns, SHA-256 a51d6d791e034922fdf1751ba24e98f9c1ad3997f61f968700f7107bc1acada6.

Current-estimation evidence. The accompanying Dyness Passive Balancing Current Estimation Technical Note documents the matched-cell differential method used to estimate the effective bleed resistance and selected-channel current. The four anchor samples used in that note are reproduced in the supplied calculation CSV. The note is preserved unchanged as an accompanying evidence document; its SHA-256 is f1e9327e8f3ad1a5a14037d4d42810438050990ef6c77736a4b3d4a9f2005c7e.

16. Final threshold reference

Function / parameter	**Observed criterion**	**Observed result**
Battery software version	2.50-71.10.11	Version under test
Balancer global enable	Ibat >= 1.5 A for 30 s	Passive balancing enabled; selected channels approx. 60 mA each
Balancer global disable	Ibat <= 1.4 A for 30 s	Passive balancing disabled
Individual resistor ON	Vcell - Vmin >= 30 mV	Resistor for that cell enabled
Full detection	Vmax >= 3.5 V	Full status; SOC = 100%; CCL = 0 A; Charge Enabled = false
Normal charge permission restored	SOC 100% -> 99%	CCL = 56 A; Charge Enabled = true
Charge MOSFET OFF	Vmax >= 3.6 V	Physical charging path interrupted

Function / parameter	Observed criterion	Observed result
Charge MOSFET ON during discharge after OVP	Discharge detected after cutoff	Physical charging path restored
Balancing at 100% SOC	Actual Ibat >= 1.5 A and cell delta criterion met	Balancing continues despite CCL = 0 A; approx. 60 mA per selected cell
Explicit balancing flag	None identified	No change in decoded status bytes during ON/OFF transitions
Full/high-cell flag	status3: 0x80 -> 0x88	Bit 0x08 set

Operational rule: The external charger should always obey CCL. The 3.6 V charge-MOSFET cutoff is an emergency protection threshold, not a permissible charging target.